FS Exam Prep — Phase 1 Spec: First Build, Signing & Device Smoke Test

Version · v1.0.1 — 2026-10-24 (v1.0.0 same day; revised: applicationId decided — W3 is now an apply step, not a gate)

Base state · Repo `crieck2010/FS-Exam-Prep-Mobile-App` @ `51c7823` (Phase 0 audited PASS 3/3; GUI §4 closed 10/10)

Where · Charlie's Windows PC — Android Studio + terminal + his Android phone. This phase is **local**; Google AI Studio cannot do it. The companion prompt targets the agent working on his PC.

Source · Launch Roadmap v1.0.0, Phase 1

Purpose

Produce the first verified release build of the app and prove it runs on a real device. Every audit to date is static (code reading); nothing has ever been compiled or run by us. This phase closes that gap and creates the signing setup every later phase depends on.

Work items (in order — W1 before any key exists)

W1 — Repo safety first. The repo's `.gitignore` covers `debug.keystore` and `.env` but **not** `*.jks`, `*.keystore`, `my-upload-key.jks`, or `keystore.properties`. The Gradle release config defaults its keystore path to `${rootDir}/my-upload-key.jks` — a key created at the project root today would be one careless commit away from being pushed to GitHub. Add the patterns, then verify with `git check-ignore` / `git status` that a test keystore filename is ignored. Commit this change on its own.

W2 — Version adoption. Set `versionName "0.9.0"` and `versionCode 2` in `app/build.gradle.kts` (currently `1.0` / `1`). Nothing else in the build file changes — in particular, do **not** edit the `signingConfigs` blocks (AI Studio guardrail comments in that file; signing is handled via the wizard/env route in W4).

W3 — applicationId: DECIDED 2026-10-24. Final value: `com.crieck.prep.fs` — family pattern `com.crieck.prep.*` (PS `.ps`, FE

`.fe`, PE `.pe`, state PLS `.pls.<state>`). (An earlier pick, `com.benchmarkprep.fs`, was switched away the same day after a collision check: benchmarkprep.com is an active SAT/ACT prep company and BenchPrep is a major exam-prep platform partnering with NCEES.) The agent changes ONLY the `applicationId` line in `defaultConfig`, as its own commit. The `namespace` stays `com.example` — it is the code package, not the store identity, and changing it breaks the build. Nothing may be uploaded to Play under the old `com.aistudio…` ID.

W4 — Upload key (Charlie's hands only). Charlie creates the key himself in Android Studio (Build → Generate Signed App Bundle/APK → Create new…), alias `upload` (matches the existing Gradle convention). Passwords are typed by Charlie and never shown to, stored by, or logged by the agent. Keystore file lives outside the repo where possible; if it must sit at the project root (`my-upload-key.jks`), W1 must already cover it. Charlie backs up the .jks + passwords to his password manager plus one offline copy before proceeding.

W5 — Build. With the W4 key wired (Android Studio wizard or the `KEYSTORE_PATH` / `STORE_PASSWORD` / `KEY_PASSWORD` environment route the Gradle config already reads): `./gradlew bundleRelease` (the future upload artifact) and `./gradlew assembleRelease` (the device-install APK). Report the outcome and any warnings; a failed build stops the phase.

W6 — Device smoke test. Install the release APK on Charlie's phone (adb or Studio), fresh (uninstall any prior copy first), and walk the script in §Smoke. Capture a screenshot at each step to a `smoke-test-0.9.0/` folder **outside** the repo, and record logcat for the whole session.

W7 — Report. Per-step pass/fail against §Smoke, the screenshot set, any crash stack traces verbatim, artifact paths, and the exact version/applicationId values built. Failures are reported, **not fixed** — app-code fixes route back through a Studio fix prompt + Muse re-audit (builder-of-record discipline).

§Smoke — the script (all steps, in order)

- **S1 Fresh launch.** First screen is Today; greeting shows the handle; readiness card, streak, daily goal render; app is in LIGHT mode if the phone is (no forced dark).

- **S2 Theme behavior.** Flip the phone to dark → app follows. Open the gear → pick Light explicitly → app stays light. Return to System → follows the phone again.
- **S3 Today.** Exactly: greeting, readiness card, ONE primary button, Up Next row. Primary button starts a quiz.
- **S4 Domain quiz.** From Practice: all 7 domain rows listed. Start one; answer 3 questions (one deliberately wrong); the explanation appears; finish to the score summary. No "NCEES" or cut-score text anywhere (the §2.3 disclaimer is the only allowed occurrence, and it lives on Progress).
- **S5 Adaptive.** Start Adaptive Practice from Practice; it launches a drill.
- **S6 Mock exam.** Practice card reads "Timed 5h 20m" and "110 exam questions covering all 7 FS domains". Start it: countdown runs from 5:20:00, question palette opens, flagging works. Exit mid-exam (finishing 110 questions is NOT required).
- **S7 Learn.** Study Guide opens with serif headings; Flashcards open and a card flips.
- **S8 Progress.** Badges row shows counts; leaderboard row is titled "Sample Leaderboard"; the leaderboard screen shows the sample note line; diagnostics list all 7 domains; the S4 session appears in the history list; the non-affiliation disclaimer sits at the bottom, verbatim.
- **S9 Quiz Configuration.** From Practice, configure a 5-question custom quiz and run it to completion.
- **S10 Persistence.** Force-stop the app and relaunch: streak, daily-goal progress, and session history from S4/S9 are intact.
- **S11 Crash scan.** The session logcat contains no `FATAL EXCEPTION` for the app package.

Phase gate: build succeeds; S1–S11 pass (or failures are documented and routed to a fix prompt); artifacts + report filed.

Non-goals

No feature work · no dependency changes · no theme/token changes · no Play Console actions (Phase 2) · no applicationId change without Charlie's explicit decision (W3).